EP 06 Young's Modulus

1. Objective

1. To familiarize with the use of a micrometer caliper and a reading microscope.
2. To master the graphic method for data processing and measure the Young's modulus of steel.

2. Principle

2.1 Core Concepts

- **Stress**: Force per unit cross-sectional area ($\text{Stress} = \frac{F}{S}$, unit: Pa), where F is tensile force and S is wire cross-sectional area.
- **Strain**: Relative elongation ($\text{Strain} = \frac{\Delta l}{l}$, dimensionless), where Δl is wire elongation and l is original length.
- **Young's Modulus (**E**)**: Stiffness indicator of materials, calculated as $E = \frac{\text{Stress}}{\text{Strain}} = \frac{F \cdot l}{S \cdot \Delta l}$.

2.2 Experimental Logic & Linear Relationship

- **Force & Area Calculation**: Tensile force $F = Mg$ (M = weight mass, $g =$ gravitational acceleration); wire cross-sectional area $S = \frac{1}{4}\pi d^2$ (d = wire diameter).
- **Elongation Measurement**: Wire elongation Δl equals the downward displacement of the cross board (attached to the wire end), measured via a reading microscope.
- **Linear Graphic Method**: Rearrange E formula to $\Delta l = \left(\frac{4gl}{\pi d^2 E}\right) M$ — Δl is linearly proportional to M. Plot $\Delta l - M$ graph, get slope $k = \frac{4gl}{\pi d^2 E}$, then solve $E = \frac{4gl}{\pi d^2 k}$.

3. Procedure

1. **Measure Wire Diameter**: Use micrometer to measure d at 5 positions, calculate average $\bar{d}$ (correct with zero reading d_0).
2. **Level Stand**: Adjust base screws to level the stand; ensure frame is parallel to the wire, and microscope scale aligns with cross board’s "|".
3. **Calibrate Microscope**: Adjust eyepiece for clear scale; lock base when cross board is visible; align reticule with cross’s "-", record initial reading C_0.
4. **Load & Record**: Add 9 weights (0.200 Kg each), record microscope readings C_i ($i = 1 - 9$) after each load.
5. **Unload & Record**: Remove weights step-by-step, record C_i' ($i = 8 - 0$); calculate average

$\bar{C}_i = \frac{C_i+C_i'}{2}$.

6. **Calculate Elongation**: $\Delta l_i = \bar{C}_i - \bar{C}_0$ ($i = 0 - 8$).

4. Data Recording & Processing

4.1 Data Form

micrometer caliper: zero reading $d_0(m)=$________

$d(m)$	1	2	3	4	5	average

m_i(Kg)		$c_i(load)$ ($\times 10^{-3}m$)		$c'_i(unload)$ ($\times 10^{-3}m$)		$\bar{c}=(c_i+c'_i)/2$ ($\times 10^{-3}m$)		$\Delta l=\bar{c}_i-\bar{c}_0$	
m_0		c_0		c'_0		$\bar{c}_0$			
m_1		c_1		c'_1		$\bar{c}_1$		Δl_1	
m_2		c_2		c'_2		$\bar{c}_2$		Δl_2	
m_3		c_3		c'_3		$\bar{c}_3$		Δl_3	
m_4		c_4		c'_4		$\bar{c}_4$		Δl_4	
m_5		c_5		c'_5		$\bar{c}_5$		Δl_5	
m_6		c_6		c'_6		$\bar{c}_6$		Δl_6	
m_7		c_7		c'_7		$\bar{c}_7$		Δl_7	
m_8		c_8		c'_8		$\bar{c}_8$		Δl_8	
m_9		c_9							

4.2 Core Calculation

Step 1: Calculate Slope k

Plot $\Delta l - M$ graph, draw the best-fit line. Select two distant points (e.g., $(M_1, \Delta l_1)$ and $(M_8, \Delta l_8)$), compute $k = \frac{\Delta l_8 - \Delta l_1}{M_8 - M_1}$ (unit: $\mathrm{m/Kg}$).

Step 2: Compute Young's Modulus E

Substitute $g =$, l, $\bar{d}$, k into:$E = \frac{4gl}{\pi \bar{d}^2 k}$

Step 3: Uncertainty Analysis

- **Main Error Sources**:

- **Diameter uncertainty (**u_d**)**: From micrometer precision (±0.00001 m) and wire inhomogeneity, estimate $u_d \approx 0.00002\ \mathrm{m}$.
- **Elongation uncertainty (**$u_{\Delta l}$**)**: From microscope precision (±0.0001 m) and elastic hysteresis, estimate $u_{\Delta l} \approx 0.0002\ \mathrm{m}$.

- **Slope uncertainty (**u_k**)**: From $\Delta l - M$ data dispersion, estimate $u_k \approx 0.0001 \mathrm{~m/Kg}$.
- **Total Uncertainty (**u_E**)**:
  Synthesize key uncertainties, approximate u_E
- **Final Result:** $E =$

5. Error Analysis

- **Systematic Errors**:
  1. Micrometer zero error: Correct with d_0;
  2. Parallax error: Align eye perpendicularly to microscope scale;
  3. Elastic hysteresis: Average loading-unloading readings.
- **Random Errors**:
  1. Wire diameter inhomogeneity: 5-point measurement;
  2. Weight mass deviation: Use calibrated weights;
  3. Slope fitting error: More data points + best-fit line.

6. Questions

1. Why average loading and unloading readings?
   Metal wires exhibit elastic hysteresis. They don't immediately return to their original length when unloaded, causing differences between loading and unloading readings. Averaging these values reduces this systematic error, making $\bar{C}_i$ closer to the true position.
2. How does a bent wire affect results?
   A bent wire makes the measured original length l larger than the actual effective length (since the bent part doesn't stretch). This overestimation of l leads to a larger calculated E when substituted into the formula.